

Subject-Calibrated EOG-Guided Diffusion for EEG Ocular Artifact Removal

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Ocular propagation into electroencephalography (EEG) varies across people, making a population contamination model imperfect for a new user. We study a narrow deployment setting in which an early same-session EEG+EOG calibration estimates a subject-specific EOG-to-EEG operator and conditions a fixed residual diffusion denoiser for later EEG+EOG query windows. Participant identity, query clean targets, labels, and outcomes are excluded from inference. On all nine BCI Competition IV-2b development participants, matching calibration reduces paired reconstruction RRMSE relative to population conditioning by $U_P = 0.03473$ and relative to a frozen cyclic unseen-subject operator by $U_W = 0.04836$. Both effects are positive for 9/9 participants after averaging three training seeds (one-sided exact sign-flip $p = 0.001953$). Mean natural EEG proxies show EOG attenuation of 0.1043 and preservation of 0.7982, with participant-level exceptions disclosed. The evidence is three-seed stability on one development cohort, not confirmation. The method is EOG-guided rather than EEG-only, and diffusion does not outperform the matched deterministic or linear estimators (RRMSE 0.11350, 0.11225, and 0.08760, respectively). The supported conclusion is therefore limited to the benefit of matching subject calibration within this fixed diffusion pipeline.

CCS Concepts: • **Computing methodologies** → **Neural networks**; *Machine learning approaches*; • **Applied computing** → *Health informatics*.

Additional Key Words and Phrases: EEG denoising, ocular artifact removal, EOG-guided denoising, subject calibration, diffusion model, brain–computer interface

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1 Introduction

Electroencephalography (EEG) recorded near the eyes contains activity propagated from blinks and eye movements. The propagation depends on anatomy, electrode placement, impedance, and reference, so a population correction can be poorly matched to a new participant. Electrooculography (EOG) provides a direct measurement of ocular activity, but the transfer from EOG to EEG remains participant dependent. This motivates a narrowly scoped adaptation question: does an early EEG+EOG calibration improve a fixed diffusion denoiser on later EEG from the same participant?

We study this question under a disjoint early-support to later-query protocol. The support segment estimates an EOG-to-EEG contamination operator without using participant identity. During later inference, both EEG and EOG are observed. The method is therefore *EOG-guided*, not

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EEG-only. Clean targets, motor-imagery labels, and evaluator annotations are not visible to the model. The matched operator supplies a linear artifact anchor to a deterministic residual network and a conditional diffusion network. Population and mismatched-subject operators can replace the matched operator while weights, query windows, and random streams remain fixed. This isolates the effect of the support-derived subject context inside one frozen denoising pipeline.

The experiment uses BCI Competition IV-2b and covers all nine development participants. An early same-session calibration segment precedes later query windows. We report paired real-EEG/EOG-backed semi-simulation for waveform efficacy and natural motor-imagery EEG for artifact attenuation and neural-signal proxies. The three training seeds are stability repeats on the same cohort, not independent scientific samples or confirmation.

The result is specific but consistent: participant-averaged RRMSE improves by 0.03473 relative to population conditioning and by 0.04836 relative to a frozen cyclic unseen-subject operator. Both effects are positive for 9/9 participants. The result does not show that diffusion is the strongest estimator: DIFF-MATCH has RRMSE 0.11350, versus 0.11225 for DET-MATCH and 0.08760 for LINEAR-MATCH. We therefore claim a matching-subject operator effect within the fixed diffusion pipeline, rather than diffusion-specific personalization.

The contributions are:

- a disjoint same-session protocol in which early EEG+EOG calibration conditions later EOG-guided EEG denoising without participant identifiers;
- integration of a support-estimated contamination operator into a fixed deterministic-plus-diffusion residual estimator;
- direct MATCH, population, frozen cyclic unseen-WRONG, and seen-donor sensitivity comparisons under common queries and random streams;
- participant-level evidence across nine development participants and three training seeds, with paired waveform metrics and natural motor-imagery preservation proxies reported separately.

2 Related Work

2.1 Ocular-Artifact Correction

Classical approaches include EOG regression, blind source separation, and artifact subspace reconstruction. Regression explicitly maps EOG into EEG and is attractive when EOG remains available at deployment, although transfer coefficients may change across participants and sessions. ICA separates latent components but requires sufficiently stable decomposition and a component selection rule. SGEYESUB provides a careful participant-level evaluation of ocular correction and illustrates the need to measure both attenuation and neural preservation [3].

Learned denoisers instead infer a correction from population data. Paired benchmarks such as EEGdenoiseNet enabled controlled comparisons of convolutional, recurrent, and generative estimators [6]. These methods establish method-level denoising efficacy, but do not by themselves test whether a disjoint calibration from an unseen participant adds value.

2.2 Diffusion Denoising

Diffusion models learn a reverse process from progressively corrupted states [1]. EEGDfus and D4PM apply diffusion ideas to EEG artifact removal under paired or mixture assumptions [2, 4]. Their objectives, conditioning information, and benchmark protocols differ from ours. We do not treat those differences as shortcomings: they answer whether a denoising method works on their target benchmarks, whereas our experiment isolates matching, population, and mismatched-subject contexts inside one frozen EOG-guided diffusion pipeline.

2.3 Calibration and Subject Adaptation

EEG variability motivates alignment, normalization, domain adaptation, and participant-conditioned representations. A known participant identifier or a closed-set embedding is not directly available for an unseen user. Our setting instead derives the context from early synchronized EEG+EOG and applies it to later data. The claim is correspondingly narrow: matching support improves this fixed estimator relative to two operator controls. We do not claim that calibration is unique to diffusion, and we retain linear and deterministic comparators to expose that boundary.

3 Subject-Calibrated EOG-Guided Residual Diffusion

3.1 Setting and Information Boundary

Let s denote a held-out participant and let S_s be an early same-session calibration segment containing synchronized EEG and EOG. A later query window contains observed EEG $\mathbf{y} \in \mathbb{R}^{C \times T}$ and EOG $\mathbf{e} \in \mathbb{R}^{E \times T}$. Both are deployment inputs. Participant identity, query clean EEG, motor-imagery labels, artifact annotations, and a query-fitted transfer are unavailable to the denoiser.

For paired training and evaluation only, we write

$$\mathbf{y} = \mathbf{x} + \mathbf{a}, \quad (1)$$

where $\mathbf{x}$ is a low-ocular-activity EEG carrier and $\mathbf{a}$ is an ocular component constructed from disjoint real EEG/EOG time ranges. This is real-signal-backed semi-simulation, not natural clean ground truth. Natural EEG is evaluated separately.

3.2 Support-Derived Contamination Operators

After robust, outer-training normalization, ridge regression on S_s estimates the matched transfer

$$\mathbf{H}_s = \mathbf{Y}_s \mathbf{E}_s^\top (\mathbf{E}_s \mathbf{E}_s^\top + \lambda \mathbf{I})^{-1}, \quad (2)$$

where $\mathbf{Y}_s$ and $\mathbf{E}_s$ concatenate support EEG and EOG samples. The ridge setting and normalization are fixed without query outcomes. The population operator $\mathbf{H}_0$ is an equal-subject estimate from the seven outer-training participants. The frozen two-heldout split excludes both the recipient and its cyclic wrong donor; consequently this is not a standard eight-subject leave-one-out population model. The primary wrong control $\mathbf{H}_d$ comes from that compatible cyclic donor, which is unseen during population training.

For a context $c \in \{s, 0, d\}$, the linear artifact anchor and output are

$$\mathbf{a}_{0,c} = \mathbf{H}_c \mathbf{e}, \quad \widehat{\mathbf{x}}_{\text{lin},c} = \mathbf{y} - \gamma_s \mathbf{a}_{0,c}. \quad (3)$$

The scalar γ_s is fixed from support only and shared across MATCH, POP, and WRONG. Thus the operator comparison does not use query outcomes to tune correction strength.

3.3 Deterministic Residual Estimator

The linear operator is imperfect. A one-step network f_ϕ predicts the normalized remaining artifact

$$\widehat{\mathbf{r}}_{\text{det},c} = f_\phi(\mathbf{y}, \mathbf{e}, \mathbf{a}_{0,c}). \quad (4)$$

Let σ be a robust per-channel scale estimated on outer-training artifact residuals. The deterministic correction is

$$\widehat{\mathbf{a}}_{\text{det},c} = \mathbf{a}_{0,c} + \sigma \odot \widehat{\mathbf{r}}_{\text{det},c}. \quad (5)$$

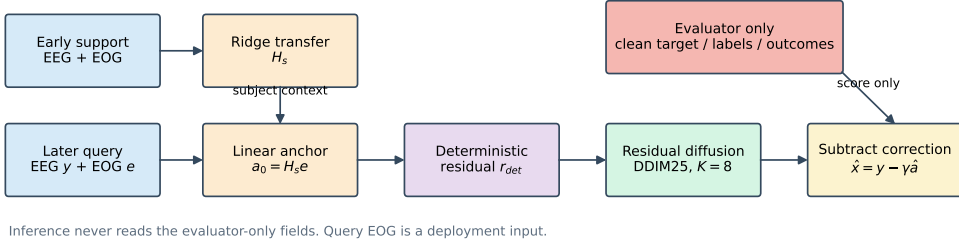


Fig. 1. Same-session subject-calibrated, EOG-guided protocol. Early support EEG+EOG estimates H_s . Later query EEG and EOG produce the linear anchor $a_0 = H_s e$; deterministic and diffusion residuals complete the correction. Clean targets, labels, and natural-query evaluator annotations are isolated from inference.

3.4 Diffusion of the Remaining Artifact Residual

Diffusion models only the normalized artifact residual left after the linear and deterministic estimates. On a paired training row, its target is

$$\delta_c^* = (a - a_{0,c}) \oslash \sigma - \widehat{r}_{\text{det},c}. \quad (6)$$

Under a cosine schedule, the forward process is

$$\delta_t = \sqrt{\alpha_t} \delta_c^* + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (7)$$

A one-dimensional U-Net predicts the v parameterization conditioned on $(y, e, a_{0,c}, \widehat{r}_{\text{det},c})$. We use EMA weights, DDIM with 25 reverse steps, and $K = 8$ posterior streams. The point estimate is their arithmetic mean, $\overline{\delta}_c = K^{-1} \sum_k \widehat{\delta}_c^{(k)}$; no sample is selected by an outcome. The complete diffusion correction and reconstructed EEG are

$$\widehat{a}_{\text{diff},c} = a_{0,c} + \sigma \odot (\widehat{r}_{\text{det},c} + \overline{\delta}_c), \quad (8)$$

$$\widehat{x}_{\text{diff},c} = y - \gamma_s \widehat{a}_{\text{diff},c}. \quad (9)$$

3.5 Context Interventions and Randomness

DIFF-MATCH, DIFF-POP, and DIFF-WRONG differ only through H_c and the quantities downstream of it. They share model weights, normalization, query, γ_s , and the window-level $K = 8$ noise bank. The cyclic unseen-WRONG is the primary specificity control. Compatible donors that were seen during population training are reported separately as a donor sensitivity analysis; they are not additional unseen replications. A temporal-shuffle control destroys the order of query EOG while retaining its amplitude distribution.

4 Experiments

4.1 Data, Protocol, and Comparisons

We use BCI Competition IV-2b [5], which contains three EEG and three EOG channels sampled at 250 Hz. All nine participants are development participants. For each participant and each of three frozen same-session units, early EEG+EOG forms support and later EEG+EOG forms query; support and query time ranges do not overlap. Table 1 summarizes the evidence unit and information boundary. Query EOG is an explicit deployment input.

Paired rows combine real low-EOG EEG carriers and real EOG-derived ocular components from nonoverlapping time ranges. We report RRMSE, waveform correlation, ΔSNR , and 1–45 Hz

Table 1. Frozen same-session development protocol.

Item	Frozen value
Participants	9 development participants
Scientific unit	Participant ($n = 9$)
Within-participant units	3 same-session units
Support	Early synchronized EEG+EOG
Query input	Later EEG+EOG
Model seeds	20260808, 20260810, 20260811
Population training	7 subjects; recipient and cyclic donor held out
Primary sampler	DDIM25, $K = 8$ arithmetic mean
Evaluator-only	paired target; natural labels and outcome fields

paired spectral utility. These targets support controlled reconstruction comparisons, not a claim of natural clean ground truth. Natural motor-imagery EEG provides EOG attenuation, low-artifact preservation, MI-band PSD distortion, covariance distortion, ERD/ERS preservation, and a frozen downstream MI readout.

The core methods are RAW; LINEAR-MATCH from Eq. (3); DET-MATCH from Eq. (4); and the same fixed diffusion model under population, matched, and cyclic unseen-wrong operators. DIFF-TEMPORAL- SHUFFLED and compatible seen-donor operators are secondary controls. The primary subject effects are

$$U_P = R(\text{DIFF-POP}) - R(\text{DIFF-MATCH}), \quad (10)$$

$$U_W = R(\text{DIFF-WRONG}) - R(\text{DIFF-MATCH}), \quad (11)$$

where R is participant-level paired RRMSE and positive values favor matching.

4.2 Aggregation and Statistical Scope

Windows are aggregated within each same-session unit, units within participant, and then the three training seeds within participant. Thus $n = 9$; the 27 participant–seed comparisons are stability measurements, not independent scientific samples. We report the exact one-sided sign-flip test aligned with the directional hypothesis and a two-sided sensitivity test. With 9/9 signs in the same direction, these are $p = 0.001953$ and $p = 0.003906$, respectively. Participant bootstrap intervals use 20,000 resamples and are descriptive because the cohort was used throughout development.

For interpretation only, we define

$$I_P = [R(\text{DET-POP}) - R(\text{DET-MATCH})] - [R(\text{DIFF-POP}) - R(\text{DIFF-MATCH})]. \quad (12)$$

A positive I_P means the deterministic subject benefit is larger; it is not evidence of diffusion-specific synergy. No interaction sign is a study gate.

4.3 Subject-Conditioning Results

Table 2 gives three-seed method means. DIFF-MATCH improves RRMSE over DIFF-POP by $U_P = 0.03473$ (median 0.03338; descriptive 95% interval [0.02899, 0.04079]) and over the cyclic unseen-WRONG by $U_W = 0.04836$ (median 0.03862; [0.02987, 0.07220]). Both effects are positive for all nine participants (Fig. 2).

The per-seed mean U_P values are 0.03654, 0.03465, and 0.03299; mean U_W values are 0.04667, 0.04779, and 0.05062. Each seed has 9/9 positive participants for both comparisons. Figure 3 shows that the direction is not created by seed averaging.

Table 2. Three-seed development means. Lower RRMSE is better; correlation, Δ SNR, attenuation, and preservation are higher-is-better.

Method	RRMSE ↓	Corr. ↑	Δ SNR ↑	EOG attn. ↑	Preserv. ↑	Cov. dist. ↓
RAW	0.48851	0.89717	0.0000	0.0000	1.0000	0.0000
LINEAR-MATCH	0.08760	0.99304	18.0654	0.0973	0.8123	0.0909
DET-MATCH	0.11225	0.99319	12.5072	0.1039	0.7979	0.1248
DIFF-POP	0.14823	0.98818	9.9214	0.0987	0.7920	0.1241
DIFF-MATCH	0.11350	0.99289	12.3898	0.1043	0.7982	0.1237
DIFF-WRONG	0.16186	0.98415	9.6218	0.0987	0.7926	0.1237

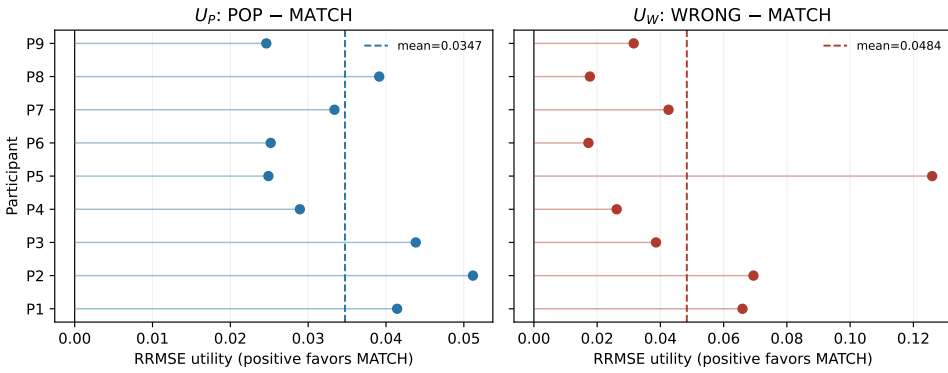


Fig. 2. Participant-level subject effects after averaging the three training seeds. Positive values indicate lower RRMSE for DIFF-MATCH. Every participant is positive for both the population (U_P) and frozen cyclic unseen-WRONG (U_W) comparison. Intervals are descriptive participant bootstrap intervals.

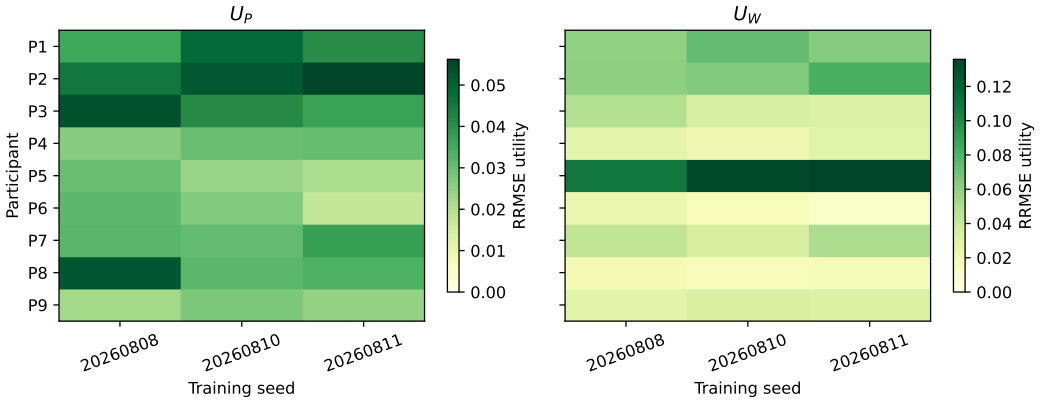


Fig. 3. Participant-by-seed effects. Training seeds are stability repeats on the same development cohort and are not independent samples.

The primary specificity control is the frozen cyclic unseen-WRONG. Each fold has one truly unseen donor. All seven compatible population-training donors were also evaluated separately as a

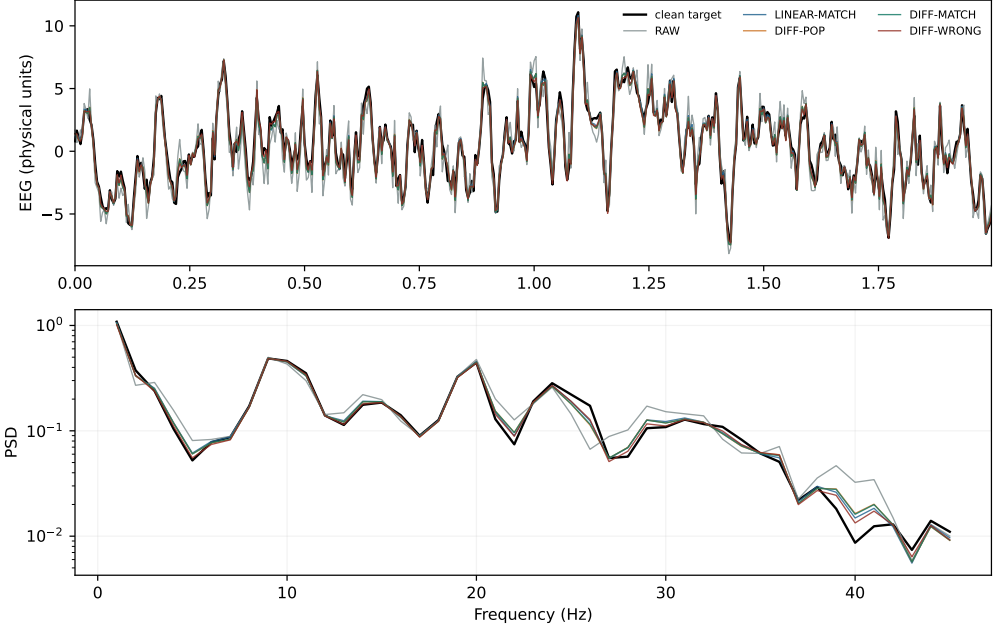


Fig. 4. Pre-frozen example: participant 1, first same-session unit, first paired window, first EEG channel, seed 20260808. The rule was fixed by indices rather than by outcome. PSD uses the valid first 500 samples.

sensitivity analysis: the recipient–seed mean utility is 0.03976 (27/27 positive). The single unseen donor sensitivity is 0.04876 (27/27 positive), but this is the same donor role as the primary cyclic control and is not a multi-unseen-donor replication.

4.4 Natural-Signal Proxies and Exceptions

Mean natural-signal proxy criteria were met, with disclosed participant-level exceptions. DIFF-MATCH has mean EOG attenuation 0.1043, preservation 0.7982, covariance distortion 0.1237, and MI-band PSD distortion 0.0719. Relative MI kappa versus DIFF-POP is +0.0032. At the participant level, EOG attenuation is nonpositive for 0/9, preservation is below 0.78 for 1/9, covariance distortion exceeds 0.15 for 2/9, and the MI-kappa margin is below -0.02 for 0/9. Accordingly, we do not claim uniform safety.

Figure 5 places participant-level EOG attenuation beside preservation and identifies the frozen safety reversals. It is a proxy analysis, not a claim that all neural activity is preserved.

4.5 Comparison Boundary and Compute

The subject effect must not be conflated with diffusion superiority. DIFF-MATCH RRMSE (0.11350) is slightly worse than DET-MATCH (0.11225) and worse than LINEAR-MATCH (0.08760). Thus diffusion-over-deterministic and diffusion-over-linear are *not supported*. The mean I_p is also positive and small across seeds, indicating that the deterministic subject benefit is slightly larger rather than showing diffusion-specific synergy.

DDIM25 with $K = 8$ uses 200 network calls, 0.0166 seconds per window on the recorded GPU runs, and a peak allocated memory of 177 MB. These resource values locate the frozen implementation and are not scientific gates.

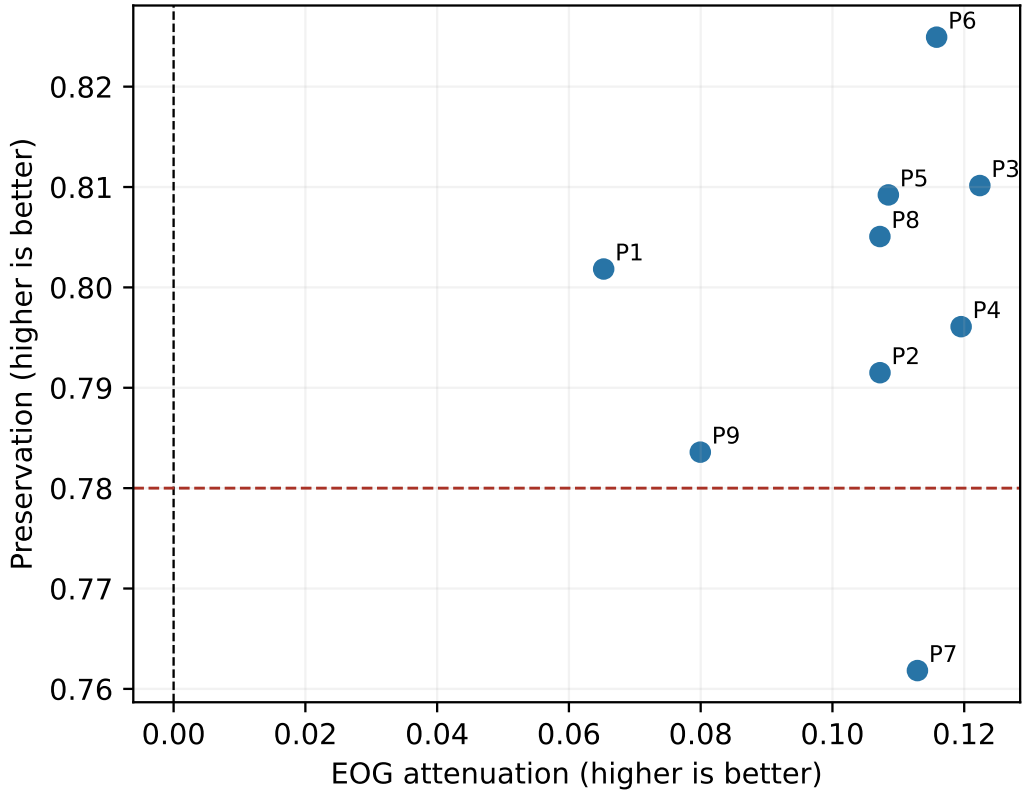


Fig. 5. Participant-level natural EEG attenuation–preservation trade-off for DIFF-MATCH. Dashed lines mark zero attenuation and preservation 0.78.

4.6 Evidence Boundary

This is three-seed stability on the same nine-participant development cohort, not evaluation on an independent cohort or confirmation. The population model uses seven training subjects because the recipient and cyclic donor are held out. The result establishes a matched-operator effect inside one EOG-guided pipeline; it does not establish EEG-only denoising, general artifact removal, benchmark leadership, or a diffusion-specific personalization effect.

5 Conclusion

Early same-session subject calibration improves a fixed EOG-guided diffusion EEG denoiser on held-out later EEG relative to population and mismatched-subject conditioning. Across nine BCI2b development participants and three training seeds, matching reduces paired RRMSE by 0.03473 versus population and by 0.04836 versus the frozen cyclic unseen-subject operator; all participants have the same direction for both effects. Mean natural-signal proxy criteria were met, with one preservation and two covariance participant exceptions.

The conclusion is deliberately narrow. Query EOG is required. The three seeds measure optimization stability on the same development cohort, not an independent-cohort evaluation. The diffusion estimator does not beat either the matched deterministic network or linear regression.

The current evidence therefore supports matching-subject conditioning within a fixed diffusion pipeline, not diffusion-specific personalization, uniform safety, or general EEG artifact removal. Independent cohorts and broader acquisition settings remain necessary.

Ethics and Privacy Statement

This study uses de-identified research EEG released by the original dataset providers; no new human-subject data were collected. The method does not learn participant identifiers or biometric embeddings. Early support EEG+EOG estimates an EOG-to-EEG contamination operator. Query EOG is an explicit deployment input, whereas query clean targets, labels, and outcomes remain evaluator-only. EEG and EOG remain sensitive physiological data and should be handled under the access and privacy terms of the source dataset.

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A Seed Stability and Donor Roles

The mean (U_p, U_w) pairs for seeds 20260808, 20260810, and 20260811 are (0.03654, 0.04667), (0.03465, 0.04779), and (0.03299, 0.05062). Each effect is positive for 9/9 participants at every seed. These 27 participant–seed comparisons are not independent; the participant-averaged effects are the inputs to the exact sign-flip tests.

For recipient $s \in \{1, \dots, 9\}$, the frozen primary wrong donor is $s + 1$ with wraparound. Both recipient and donor are excluded from population training. Each fold therefore contains one unseen wrong donor and seven donors that were seen during population training. The latter are a donor sensitivity, not unseen-donor replication.

B Safety Distribution

The participant minima/maxima complement the means in the main paper: preservation has mean/minimum 0.7982/0.7618, covariance distortion has mean/maximum 0.1237/0.1660, EOG attenuation has mean/minimum 0.1043/0.0653, and MI-band PSD distortion has mean/maximum 0.0719/0.0954. The minimum participant MI-kappa difference relative to DIFF-POP is -0.0199 . These distributions motivate reporting mean proxy criteria together with every participant-level exception.

C Reproducibility Boundary

All scientific tables are reconstructed from committed participant-level summaries. Checkpoints, prepared arrays, per-window outputs, raw data, and logs remain server-side. The evidence manifest records the source summary for every number and figure without modifying the frozen scientific CSV or JSON files.

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